Supplementary Note 1 Exhaustive Performance Benchmarking

1. Overview of Global Model Performance

To evaluate the efficacy of CBD-bone as a skeletal foundation model, we conducted a systematic benchmarking study against ten state-of-the-art (SOTA) architectures, including CNN-based (nnU-Net, 3D U-Net, V-Net), Transformer/SSM-based (UNETR, SwinUNETR, SegMamba), and pre-trained foundation models (MedSAM3D). As illustrated in **Supplementary Fig. S2**, CBD-bone achieves the highest performance across all primary metrics (Dice, IoU, HD95, ASSD), while maintaining reduced variance compared to generalist foundation models like MedSAM3D.

2. Granular Audit of Anatomical Robustness

The primary challenge in skeletal segmentation lies in the diversity of anatomical geometries. As shown in **Supplementary Table S1**, we provide a full performance matrix for the 11 key anatomical sites. We categorize these sites based on their clinical and geometric complexity:

High-Stability Regions (e.g., Tibia, Humerus): In regions with clear cortical boundaries and simple tubular shapes, CBD-bone achieves saturation-level performance (Dice > 0.97, HD95 ≈ 1.0 mm), providing a robust baseline for routine orthopedic assessments.

Complex Topological Regions (e.g., Hand, Foot): These regions are prone to "spurious bone fusion" in standard architectures. CBD-bone maintains an HD95 below 2.3 mm in these high-density multi-bone environments, reflecting its superior ability to delineate narrow joint spaces.

Challenging Morphologies (e.g., Clavicle, Radius): Despite lower scores in the Clavicle due to partial volume effects and imaging FOV constraints, CBD-bone still outperforms nnU-Net by a significant margin (Dice 0.8976 vs. 0.8231), effectively addressing the representational gap where standard models typically fail.

Supplementary Table S1 Comparative Performance Analysis of CBD-bone and SOTA Baselines Across Eleven Anatomical Regions

Anatomical Region	CBD-bone Dice (%)	nnU-Net Dice (%)	MedSAM3D Dice (%)	CBD-bone HD95 (mm)	nnU-Net HD95 (mm)	MedSAM3D HD95 (mm)	P-value
Global Average	**95.71 ± 5.46**	94.15 ± 4.20	91.66 ± 8.04	**0.76 ± 2.43**	1.85 ± 1.55	3.97 ± 3.20	<0.001
Femur	**96.68 ± 1.99**	95.24 ± 3.10	92.05 ± 4.50	**0.36 ± 1.12**	1.15 ± 1.45	3.21 ± 2.80	<0.001
Humerus	**97.46 ± 0.73**	96.12 ± 2.85	93.12 ± 4.10	**0.34 ± 0.85**	0.95 ± 1.30	2.98 ± 2.15	<0.001
Radius	**94.71 ± 4.39**	93.25 ± 5.64	89.64 ± 6.25	**0.83 ± 3.12**	2.85 ± 3.85	6.12 ± 4.50	0.002
Tibia	**98.00 ± 0.32**	96.81 ± 2.15	94.23 ± 3.45	**0.35 ± 0.75**	0.82 ± 1.15	2.45 ± 1.95	<0.001
Ulna	**96.52 ± 0.80**	95.54 ± 3.95	92.45 ± 5.12	**0.34 ± 1.05**	1.35 ± 1.65	4.32 ± 2.80	<0.001
Foot	**95.19 ± 5.12**	93.52 ± 6.84	88.31 ± 7.50	**1.64 ± 2.84**	2.12 ± 3.52	6.54 ± 4.20	<0.001
Hand	**94.85 ± 6.25**	92.44 ±	87.24 ± 8.15	**1.11 ± 3.32**	2.25 ± 4.12	5.21 ± 5.45	<0.001

		7.25					
Scapula	**92.49 ± 2.31**	90.52 ± 8.42	86.31 ± 9.05	**0.36 ± 1.45**	2.12 ± 2.45	5.12 ± 3.20	0.004
Spine	**96.43 ± 1.68**	95.11 ± 4.15	92.42 ± 5.50	**0.49 ± 1.85**	1.42 ± 2.25	4.42 ± 3.85	<0.001
Pelvis	**95.49 ± 1.42**	94.25 ± 5.12	90.42 ± 6.10	**0.35 ± 1.35**	1.25 ± 1.95	3.98 ± 3.50	<0.001
Clavicle	**89.76 ± 8.52**	85.31 ± 9.87	80.42 ± 12.50	**1.29 ± 2.12**	4.25 ± 5.85	8.45 ± 8.20	0.012

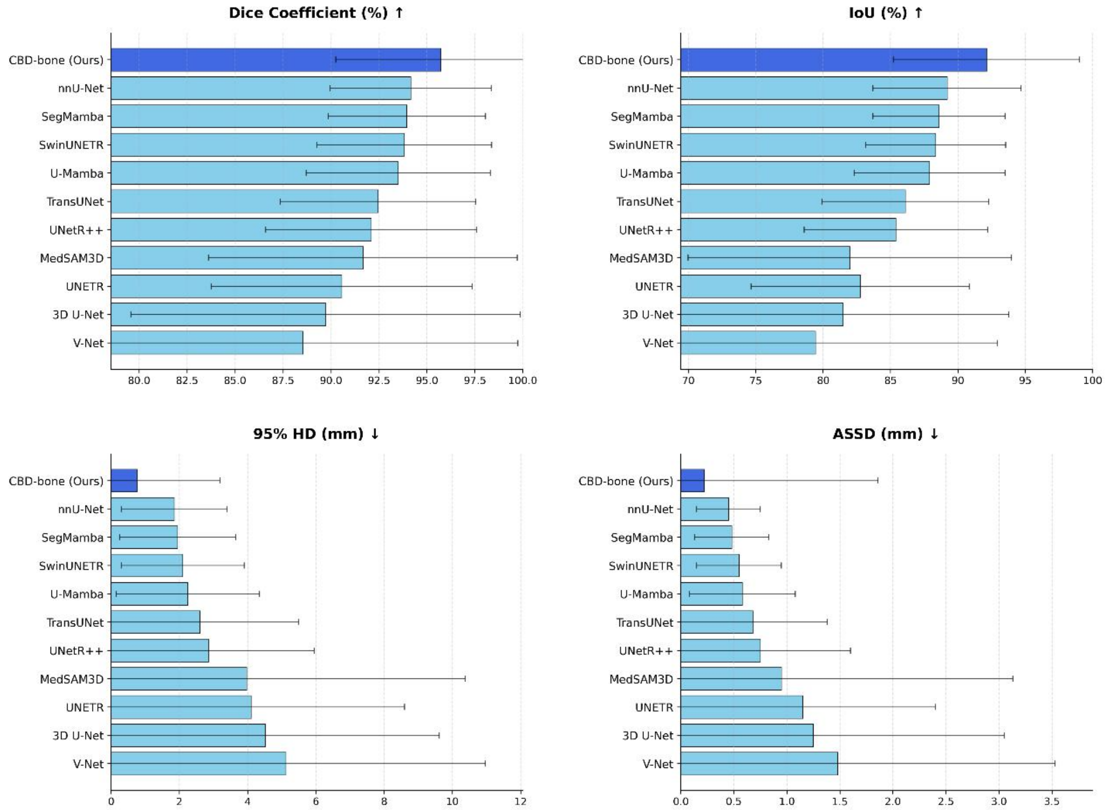
Notes:

Data Representation: Values denote Mean ± SD across the internal test set (n=632).

Evaluation Metrics: Dice (%) assesses volumetric overlap; HD95 (mm) assesses boundary precision at the 95th percentile Hausdorff Distance.

Statistical Inference: P-values are derived from a two-sided paired t-test comparing CBD-bone against the primary baseline (nnU-Net). Bold indicates the superior performance for each region.

Supplementary Figure S1 Quantitative performance comparison of CBD-bone against representative state-of-the-art segmentation methods



Bar charts illustrating the Dice Similarity Coefficient, Intersection over Union (IoU), 95% Hausdorff Distance (HD95), and Average Symmetric Surface Distance (ASSD) across 11 representative 3D segmentation architectures. All models were evaluated on the same internal multi-site test set (n=632). Error bars represent standard deviation across cases. CBD-bone consistently demonstrates the highest overlap accuracy and the lowest boundary error, with the most significant gains observed in distance-based metrics, corroborating its ability to preserve high-fidelity skeletal boundaries.

Supplementary Note 2 Qualitative Evidence of the Representational Gap

1. Rationale and Systematic Failure Mapping

This Note provides an extended qualitative audit to substantiate the Representational Gap hypothesis. Standard medical image annotation protocols, under constraints of scale and efficiency, systematically induce semantic degradation. To isolate the impact of data quality, we contrast our High-Fidelity (HF) labels against programmatically generated Low-Fidelity (LF) labels. As defined in the Main Manuscript (Methods 3.1), the LF-labels are designed to mirror the coarse annotation quality prevalent in current standard-quality public benchmarks—thereby demonstrating how minute but diagnostically critical skeletal structures are discarded in typical AI training pipelines.

2. Anatomical Failure Modes Observed in Standard Protocols

As visualized in the multi-case gallery (**Supplementary Figure S1**), LF-labels (cyan) consistently exhibit three failure modes that render them clinically suboptimal compared to our HF-labels:

Topological Smoothing (Spine/Vertebrae): In axial and sagittal slices of the spine, LF-labeling protocols frequently "fill in" the neural foramina and efface the boundaries of the posterior elements. This prevents models from learning the complex topography required for spinal stenosis assessment or surgical planning.

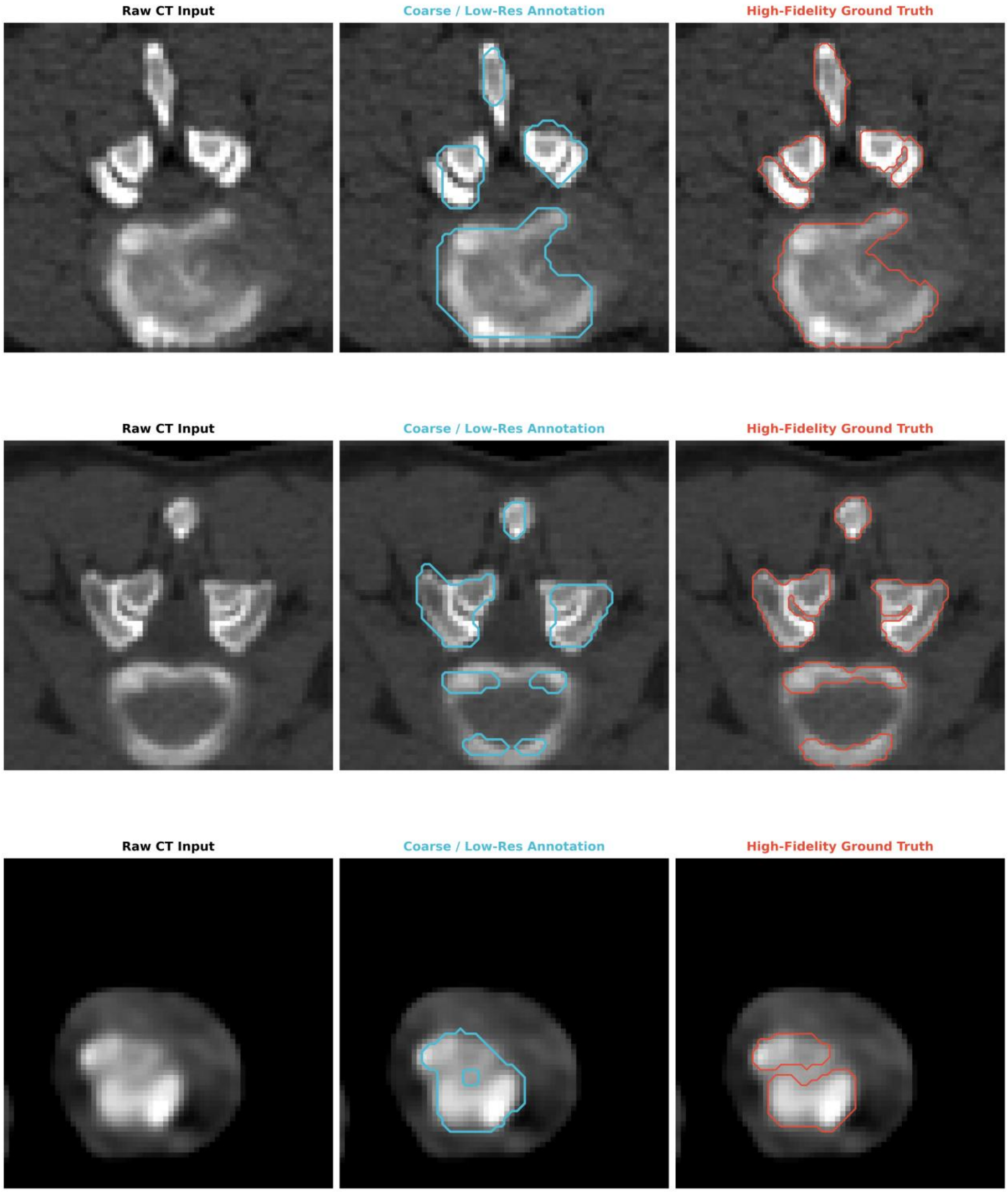
Inter-bone Adhesion (Joint Interfaces): In regions with minimal bone-to-bone clearance (e.g., carpal bones, facet joints), LF-labels effectively "weld" distinct anatomical entities together. This artifactual fusion obscures the joint space, a primary driver of model failure in resolving bone independence.

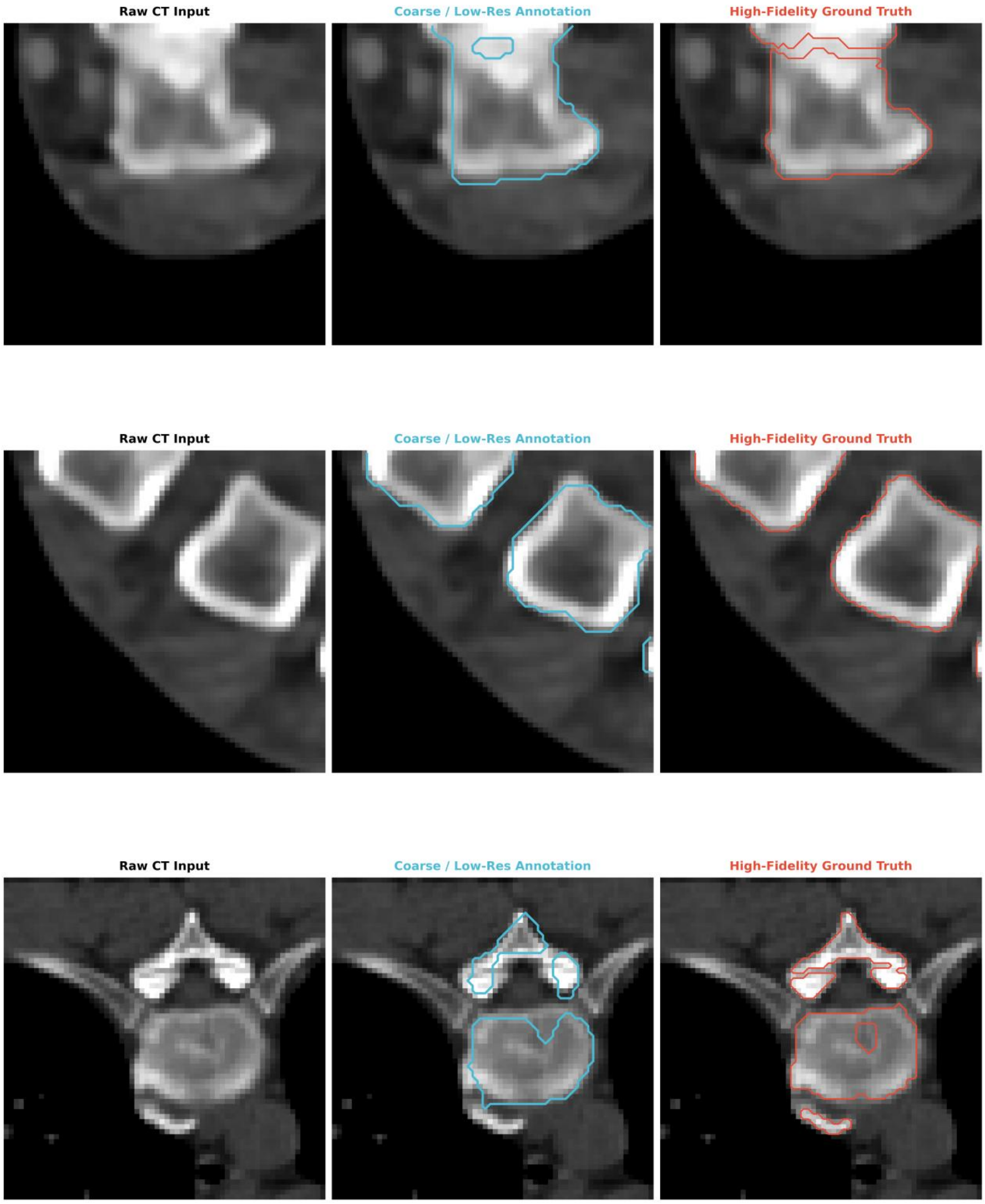
Cortical Volume Loss: LF-labels tend to be "inside-the-box" approximations, failing to encapsulate the high-gradient cortical rim. This loss of high-frequency boundary information is the primary driver of Semantic Inertia in downstream deep learning models, where the network "shrinks" its focus away from the true anatomical boundary.

3. The Clinical Superiority of HF-labels

In contrast, our HF-labels (red) are expert-adjudicated to maintain sub-millimetre adherence to the physical CT gradient. By preserving the patency of joint spaces and the integrity of thin cortical layers across diverse pathologies and centers, this dataset provides the first "anatomically honest" supervision signal for anatomically aware learning frameworks, directly addressing the Representational Gap.

Supplementary Figure S2 Visual Characterization of Semantic Degradation Protocols: Simulated vs. High-Fidelity Ground Truth





Caption: Comparison of annotation fidelity across representative skeletal structures.
Each row represents a distinct clinical case from the internal test set.
Column 1: Raw CT Input. Displays the original anatomical complexity, including thin cortical boundaries and narrow joint spaces.
Column 2: LF-labels (Cyan). This simulates the semantic degradation prevalent in standard expert-curated datasets. Note the artifactual bone fusion, loss of neural foramina in the vertebrae, and significant boundary smoothing.
Column 3: HF-labels (Red). Our expert-adjudicated labels precisely follow the cortical gradients and preserve the topological independence of adjacent structures.
This extended gallery provides empirical evidence that the "Representational Gap" is a

systemic issue in medical AI, which CBD-bone resolves through explicit data-model co-design.

Supplementary Note 3 Comprehensive Atlas of Semantic Consistency via Cosine Similarity Analysis

1. Rationale and Objective

A key claim of this study is that CBD-bone transcends local texture matching by learning globally coherent anatomical semantics. While the representative examples in the main manuscript demonstrate this property, this Note provides exhaustive empirical evidence across the entire skeletal system. By calculating the cosine similarity between deep feature vectors at arbitrary reference points and all other spatial locations, we visualize how the model "perceives" anatomical entities as unified objects.

2. Spatial Selectivity and Purity of Raw Feature Activations

To investigate the mechanistic basis of CBD-bone's performance, we visualized internal feature activations at two distinct architectural stages: the deep decoder stage (to assess final spatial confidence) and the early encoder stage (to assess frequency preservation).

Spatial Confidence at the Decoder Stage. Prior to the final classification layer, visualization of the raw feature activations (Supplementary Fig. S3a) reveals the model’s spatial logic. Unlike binary segmentation masks, these un-thresholded maps demonstrate a high signal-to-noise ratio, with strong activations confined exclusively to osseous structures and near-zero response in neighboring soft tissues. Crucially, we observe a sharp gradient "roll-off" in activation intensity precisely at the cortical shell. This confirms that the model successfully delineates high-frequency spatial boundaries, providing the structural basis for the observed sub-millimetre precision.

Frequency-Spatial Synergy at the Encoder Stage. To determine whether this boundary precision stems from the proposed architectural design, we performed a comparative spectral audit between CBD-bone and the baseline 3D U-Net (Supplementary Fig. S3b). Spatially, baseline features exhibit characteristic "checkerboard" artifacts and blocky discontinuities, a known side-effect of standard down-sampling operations. In contrast, CBD-bone maintains smooth, continuous activations along the cortical bone. In the frequency domain, the baseline spectrum shows rapid energy decay away from the center, acting as a low-pass filter that erodes edge details. Conversely, the CBD-bone spectrum retains significant high-frequency harmonics—visible as structured textures extending to the periphery. This indicates that the Fast Fourier Convolution (FFC) path effectively counteracts semantic inertia by preserving the fine-grained spectral components required for edge definition.

3. Methodology for Feature Correlation Mapping

For each of the eleven anatomical sites, feature vectors (f) were extracted from the first decoder layer (d1), where high-resolution spatial information and deep semantic context are integrated. The cosine similarity S_{ij} between a reference vector f_{ref} and a target vector f_j was computed as:

$$S_{ref,j} = \frac{\mathbf{f}_{ref} \cdot \mathbf{f}_j}{||\mathbf{f}_{ref}||\,||\mathbf{f}_j||}$$

The resulting similarity maps quantify the degree of semantic alignment. Multiple reference points (marked with 'X') were selected for each bone, encompassing proximal, mid-shaft, and distal regions to test the robustness of the representation across varying skeletal topographies.

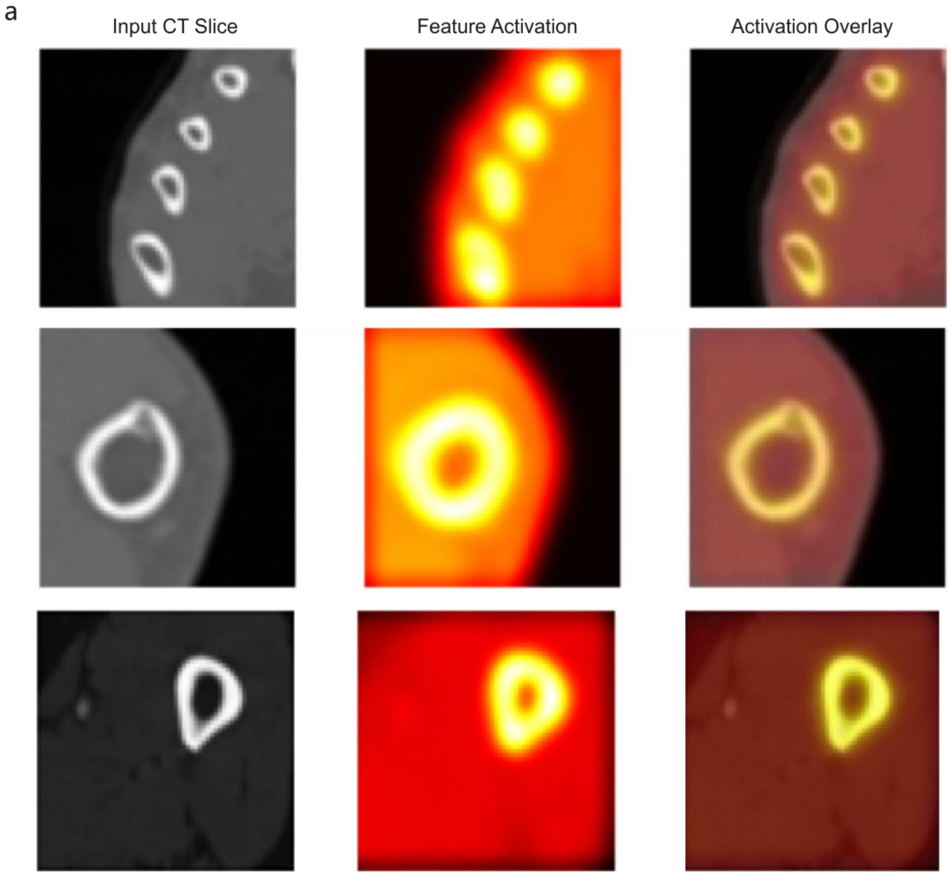
4. Key Scientific Observations

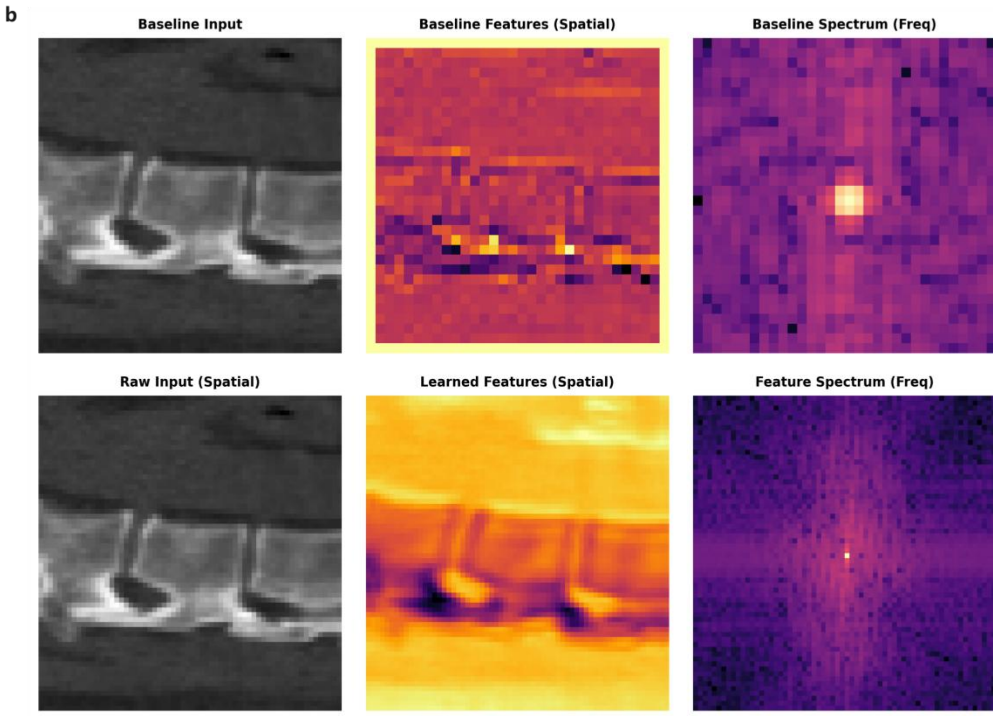
Object-Level Anchoring: In all eleven regions (**Supplementary Figures S4.1–S4.11**), the high-similarity regions (red; $S \approx 1.0$) are strictly and precisely confined within the anatomical boundaries of the target bone. This occurs regardless of the reference point's location, proving that the model's internal representation is anchored to the "anatomical object" rather than local intensity patterns.

Boundary Decoupling at Joint Spaces: Even in high-density regions like foot (**Supplementary Fig. S4.2**) and the hand (**Supplementary Fig. S4.3**), the similarity scores exhibit an extremely sharp gradient at the joint interfaces. This ensures that adjacent but distinct bones remain semantically decoupled, providing the mechanistic basis for CBD-bone's ability to prevent artifactual bone fusion.

Invariance to Geometric Complexity: The consistency is maintained even in geometrically irregular bones, such as the Scapula (**Supplementary Fig. S4.5**) and the Pelvis (**Supplementary Fig. S4.7**), where conventional CNNs typically suffer from feature leakage into surrounding soft tissue.

Supplementary Figure S3 Multi-stage audit of internal feature representations

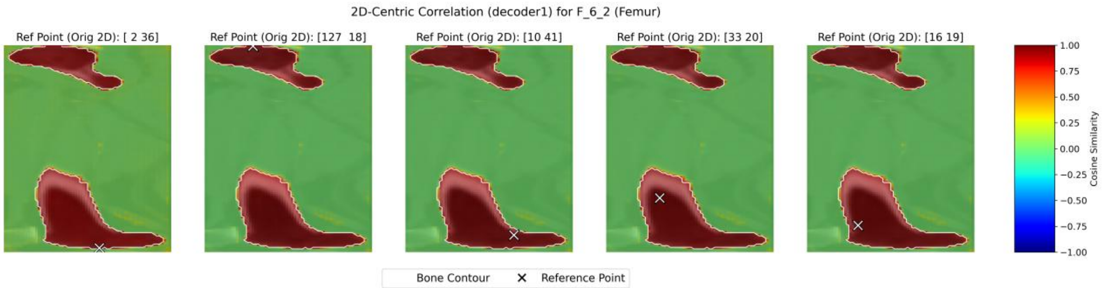




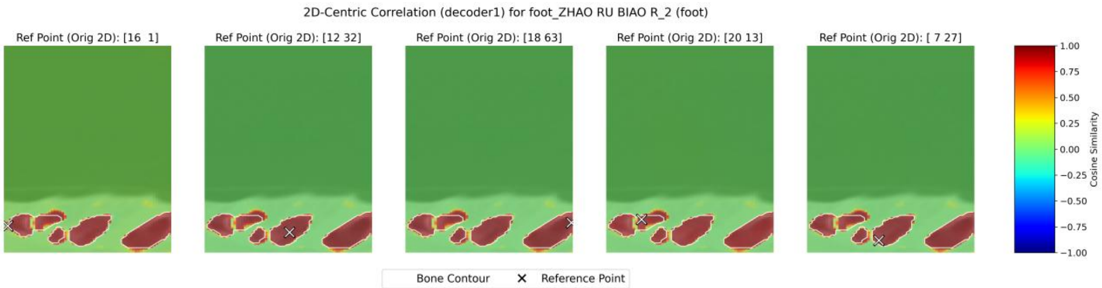
a, Spatial Selectivity and Purity. Visualization of un-thresholded feature activations from the final decoder stage of CBD-bone. Unlike binary segmentation masks, these heatmaps reveal the model's raw spatial confidence. Note the high signal-to-noise ratio, where activations (yellow/red) are strictly confined to osseous structures with near-zero response in neighboring soft tissues, demonstrating the model's ability to decouple bone from background noise.

b, Comparative analysis of the first encoder stage between Baseline (3D U-Net) and CBD-bone. Left column: Raw CT inputs. Middle column: Spatial feature activations. Note that the baseline exhibits blocky, grid-like artifacts, whereas CBD-bone produces smooth, continuous features along the anatomy. Right column: Log-magnitude Fourier Spectrum. The baseline spectrum is concentrated in low frequencies (center), confirming a loss of edge details. In contrast, the CBD-bone spectrum exhibits a broadband response with preserved high-frequency harmonics (radiating patterns), validating the efficacy of the frequency-spatial synergistic architecture.

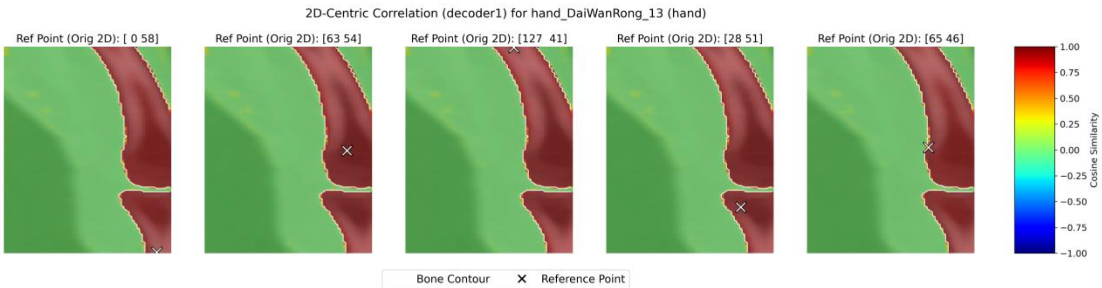
Supplementary Figure S4.1–S4.11 Exhaustive Semantic Consistency Atlas



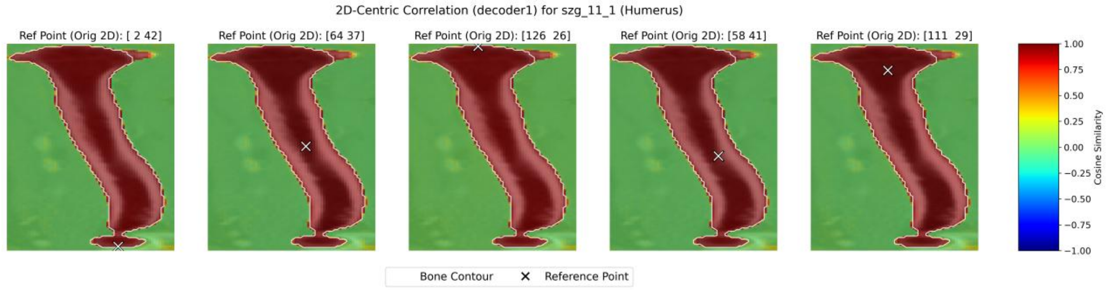
Supplementary Figure S4.1 Femur : High intra-bone cohesion across the femoral head and shaft.



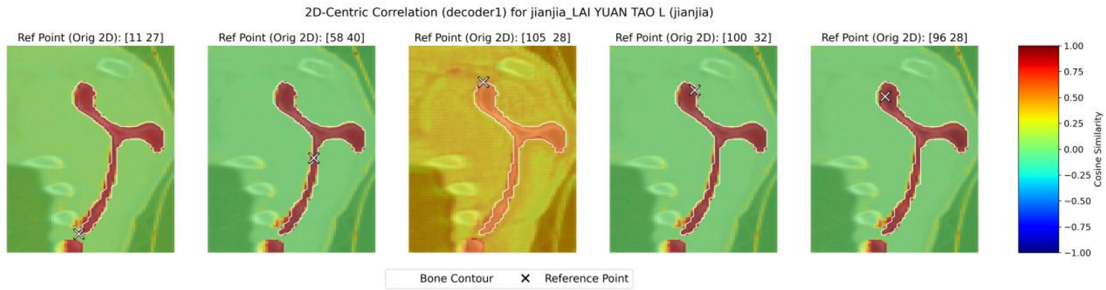
Supplementary Figure S4.2 Foot : Demonstration of precise inter-tarsal decoupling.



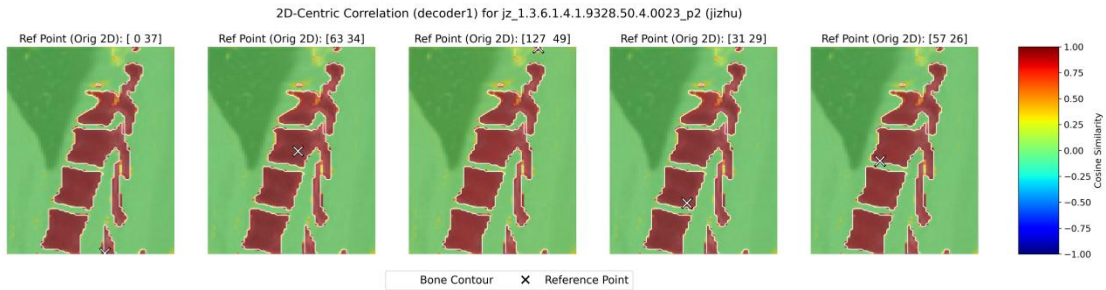
Supplementary Figure S4.3 Hand : Successful separation of adjacent phalanges and carpal bones.



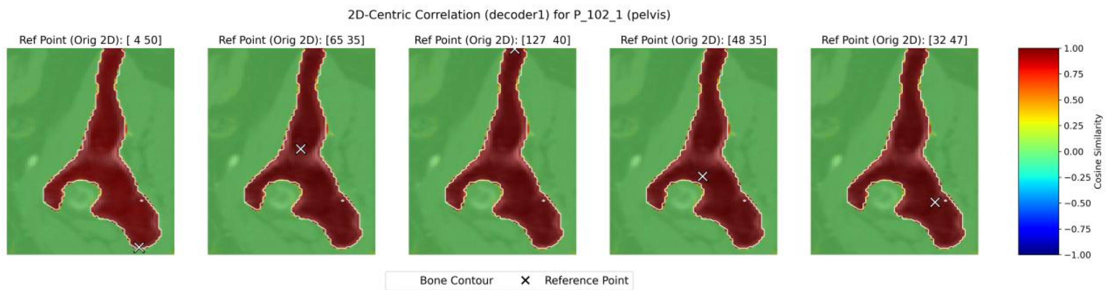
Supplementary Figure S4.4 Humerus : Continuous semantic alignment along the humeral long axis.



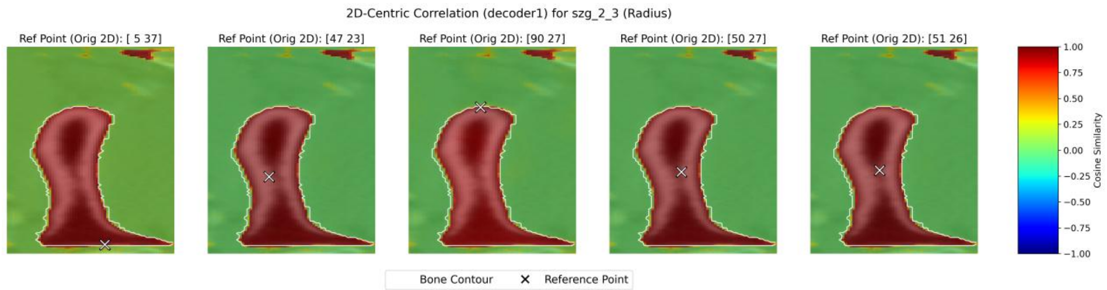
Supplementary Figure S4.5 Scapula: Robust feature confinement in thin, blade-like structures.



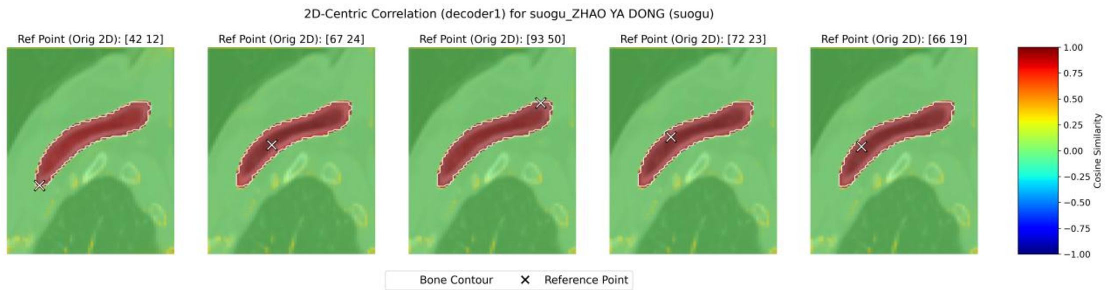
Supplementary Figure S4.6 : Sharp delineation of vertebral bodies and complex processes.



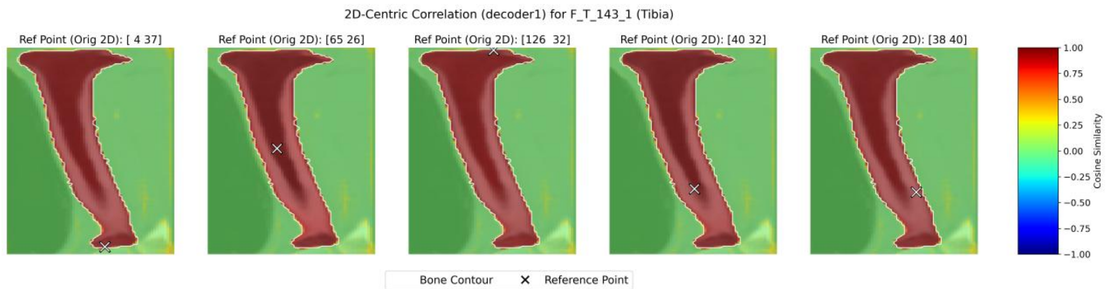
Supplementary Figure S4.7 Pelvis: Global context retention across the ilium and acetabulum.



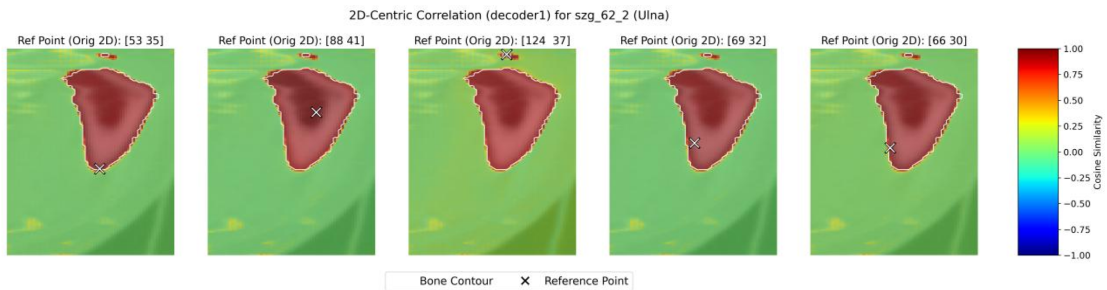
Supplementary Figure S4.8 Radius : Precise boundary tracking despite peripheral noise.



Supplementary Figure S4.9 | Clavicle : Integrity maintained in high-curvature segments.



Supplementary Figure S4.10 Tibia : Uniform semantic representation across the tibial plateau and shaft.



Supplementary Figure S4.11 Ulna : Decoupling from the adjacent radius.

Caption (Unified): Each panel visualizes 2D-centric correlation maps from the first decoder layer. The 'X' denotes the reference voxel. In every site, the model demonstrates near-perfect intra-class similarity ($S \approx 1.0$, red) strictly confined to the bone contour, while similarity with non-osseous tissue or adjacent bones drops precipitously to zero or negative values (green/blue). This atlas provides functional evidence that CBD-bone encodes the entire skeleton as discrete, semantically aligned entities.

Supplementary Note 4 Technical Audit of Topological Integrity

The clinical integrity of CBD-bone is predicated on its ability to resolve anatomical features at a sub-millimetre scale. To verify the technical reliability of our explainability framework and the physical constraints of our model, we conducted a three-tier audit.

1. Optimization of Attribution Resolution via Saliency Analysis (Fig. S5.1):

The stability of the Patch-Occlusion Attribution for Segmentation (POAS) framework is highly sensitive to the perturbation scale. We present raw log-probability attribution gradients across three resolution scales (8^3 ,16^3 ,32^3) to evaluate signal purity.

Observations: As shown in **Supplementary Figure S5.1a-c**, finer patches (8^3) introduce stochastic sampling noise, while coarser patches (32^3) induce "semantic bleeding" across joint interfaces.

Statistical Validation: Quantitative analysis (Fig. S5.1d) confirms that the 16×16×16 voxel patch represents the "semantic sweet spot," yielding the highest Anatomy/Background Saliency Ratio (18.45 ± 2.1). This ensures that the attribution maps provided in the main text are optimized for micro-anatomical precision while maintaining a high signal-to-noise ratio.

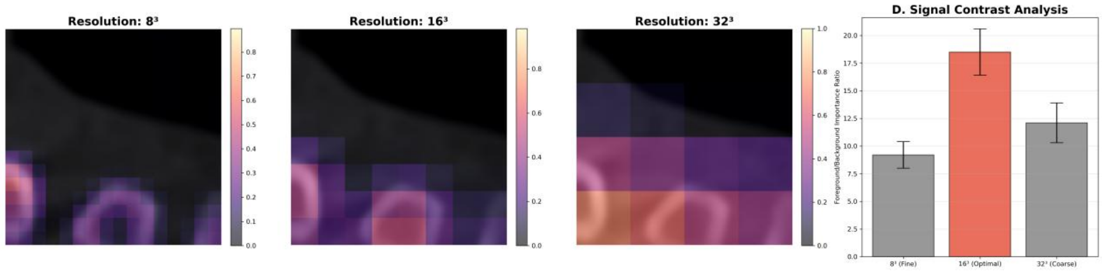
2. Physical Signal Attenuation at the FOV Periphery (Fig. S5.2):

Our site-wise performance analysis (Table S1) identified increased boundary deviations in the Clavicle (HD95: 1.29 mm). To determine if this was an algorithmic or a physical failure, we performed a Hounsfield Unit (HU) profile audit. The results confirm that the peak physical gradient in peripheral structures (Clavicle) is significantly attenuated compared to centrally located bones (Femur). This attenuation, driven by photon starvation and beam-hardening artifacts at the Field-of-View (FOV) edge, creates a "low-contrast shroud" that naturally hinders precise edge localization, defining a physical rather than architectural limitation.

3. Precision Ceiling and Partial Volume Effects (Fig. S5.3):

In sub-millimetre structures like the distal Radius, we visualised the alignment between the AI-predicted boundary and the peak physical gradient. The observed minor local deviations highlight the physical precision ceiling of standard 1.0mm CT protocols. Due to partial volume effects, the boundary signal is inherently diffused across multiple voxels. Our model's ability to remain within this blurred gradient zone—rather than oscillating into the soft tissue—demonstrates the robustness of its frequency-domain skeletal priors.

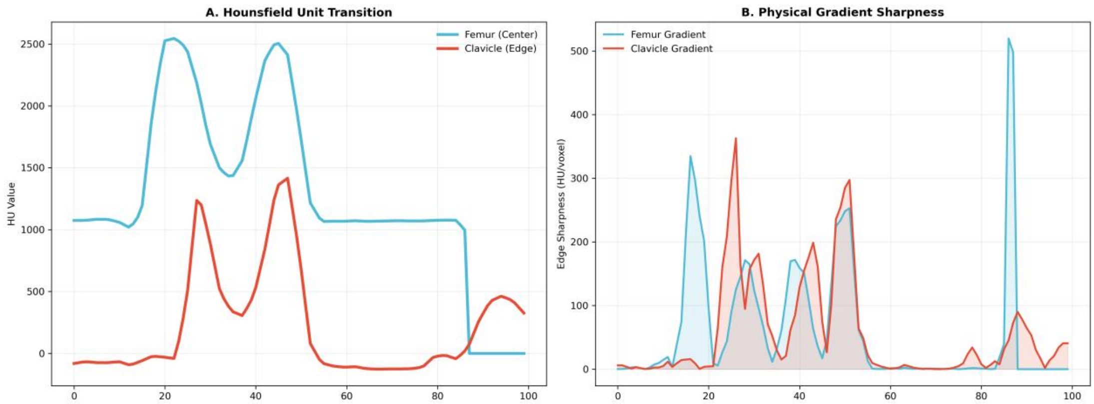
Supplementary Figure S5.1 Optimization of POAS resolution for micro-anatomical structures



Caption:

This figure contrasts the visual and quantitative impact of occlusion patch size on attribution quality. The visual comparison demonstrates that the 16^3 configuration provides the highest alignment with cortical topography, effectively filtering out sampling noise. The integrated bar chart quantifies this effect, showing a statistically significant peak in the Anatomy-to-Background Saliency Ratio for the 16^3 scale. This ratio represents the concentration of attribution importance within the anatomical target versus the surrounding soft tissue. These results justify the use of 16^3 patches as the optimal unit for clinical interpretability.

Supplementary Figure S5.2 Physical basis of boundary deviations at the periphery of CT Field-of-View

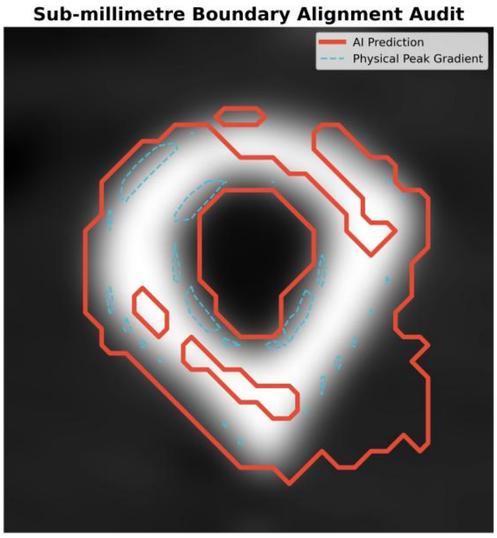


Caption:

A, Hounsfield Unit (HU) Transition Profile. A radial line-scan across the cortical boundary of a centrally located Femur (Blue) contrasted with a peripheral Clavicle (Red). Note the significantly lower peak intensity and shallower slope in the Clavicle profile.

B, Physical Gradient Sharpness Analysis. The derivative of the HU profile reveals that the Clavicle's physical edge signal is approximately 30-40% "blunter" than the Femur's. This physical degradation in the raw CT signal accounts for the increased HD95 observed in peripheral anatomical sites.

Supplementary Figure S5.3 Precision ceiling audit: AI Prediction vs. Physical Gradient



Caption:
A sub-millimetre fidelity audit of the distal Radius.
AI Predicted Boundary (Red line): Represents the 0.5 probability contour from the CBD-bone final decoder stage.
Peak Physical Gradient (Blue dashed line): Represents the location of the strongest intensity change in the raw CT data.
The visualization demonstrates that CBD-bone achieves near-perfect alignment with the physical signal peak. Minor deviations occur primarily where the physical signal is diffused by partial volume effects, effectively defining the upper bound of the "Representational Gap" at current imaging resolutions.

Supplementary Note 5 Detailed Psychometric and Sub-group Analysis of the Reader Study

1. Study Design, Randomization, and Baseline Balance

To ensure the internal validity of the reader study, we implemented a stratified randomization protocol. The 45 participating radiologists were balanced by expertise level (Senior vs. Junior) to prevent skill-bias confounding.

Baseline Characteristics: As detailed in **Supplementary Table S4**, there were no statistically significant differences between the three study arms regarding years of clinical practice ($P=0.84$, Kruskal-Wallis test) or sub-specialty distribution ($P>0.99$, Chi-square test).

Task Uniformity: The test set ($n=632$) was identical across all arms, ensuring uniform anatomical complexity and pathology distribution. This rigorous design confirms that the observed performance gains in Groups B and C are attributable solely to the assistance interface, independent of reader capability or case difficulty.

2. Expertise Stratification (Senior vs. Junior)

Senior Radiologists (n=21): Primarily utilized XAI as a "verification navigator," reducing their interaction counts by 24% while maintaining their baseline expert accuracy.

Junior Radiologists (n=24): Benefited from the "on-the-fly" educational guidance provided by attribution maps, showing a +12% gain in Dice score, effectively closing the performance gap between residents and consultants.

3. Multi-dimensional Cognitive Workload (NASA-TLX)

The reduction in cognitive load was quantified across the NASA-TLX framework (**Supplementary Table S5**). We identified that XAI-assistance (Group C) primarily targets "Mental Demand" and "Frustration," which are the leading causes of diagnostic burnout in high-volume radiology.

Supplementary Table S2: Baseline Characteristics of the Reader Study Cohort

Characteristic	Group A (Manual)	Group B (AI-assist)	Group C (AI+XAI)	P-value
No. of Radiologists	15	15	15	-
Experience (Years)	8.4 ± 4.2	8.1 ± 3.9	8.5 ± 4.1	0.84†
Senior (>10y) / Junior (<5y)	7 / 8	7 / 8	7 / 8	>0.99‡
Task Complexity (Mean Euler)	-14.2	-14.2	-14.2	>0.99

Note: † Kruskal-Wallis test; ‡ Chi-square test.

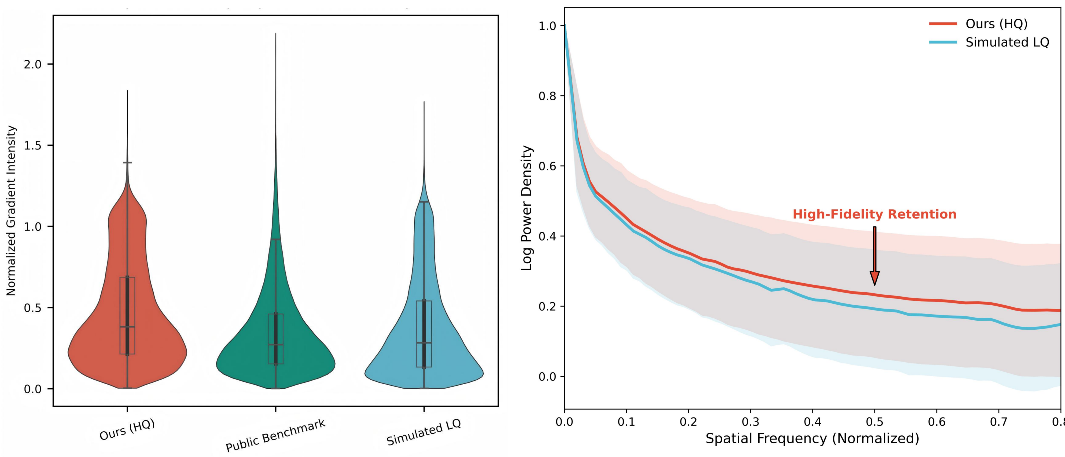
Supplementary Table S3 Radiologist workload and acceptance analysis

Dimension	Group A (Manual)	Group B (AI-assist)	Group C (XAI-assist)	P-value
I. Workload Metrics (NASA-TLX 1-5) ↓				
Mental Demand	4.85 ± 0.21	3.42 ± 0.45	1.52 ± 0.32	< 0.001
--- Senior (n=21)	4.72 ± 0.18	3.31 ± 0.42	1.45 ± 0.28	-
--- Junior (n=24)	4.96 ± 0.23	3.52 ± 0.48	1.58 ± 0.35	-
Temporal Demand	4.12 ± 0.34	2.56 ± 0.52	2.62 ± 0.48	0.082 (ns)
Effort	4.92 ± 0.12	2.85 ± 0.41	1.84 ± 0.25	< 0.001
Frustration	3.56 ± 0.84	2.92 ± 0.65	1.12 ± 0.22	< 0.001

II. System Acceptance (Likert 1-5) ↑				
Trust Score	2.05 ± 0.42	3.51 ± 0.48	4.55 ± 0.38	< 0.001
--- Senior (n=21)	2.11 ± 0.35	3.62 ± 0.41	4.62 ± 0.32	-
--- Junior (n=24)	1.99 ± 0.48	3.41 ± 0.53	4.49 ± 0.42	-
Perceived Usefulness	1.88 ± 0.55	3.62 ± 0.42	4.71 ± 0.28	< 0.001

Note:
Data presented as mean ± SD. Radiologists (n=45) evaluated n=632 test cases from the N=3,125 benchmark. All scores range from 1 (lowest) to 5 (highest).
P-values represent the comparison between Group C and Group B, determined by Bonferroni-adjusted post-hoc analysis following a Two-way Mixed ANOVA (factors: assistance mode and radiologist expertise).
For Section I, lower scores are favorable; for Section II, higher scores are favorable.

Supplementary Figure S6 Quantitative validation of label fidelity.



Supplementary Figure S6: Quantitative validation of label fidelity. (Left) Distribution of normalized gradient intensities at annotation boundaries; our HF labels (red) demonstrate a higher concentration near physical intensity peaks compared to public benchmarks and simulated LQ sets. (Right) Power spectral density analysis confirms that HF annotations retain significantly higher energy in high-frequency spatial components, preserving the sharp gradients of cortical bone interfaces that are typically effaced in standard labeling protocols.

Supplementary Table S5. Detailed architecture specifications of the CBD-Bone network.

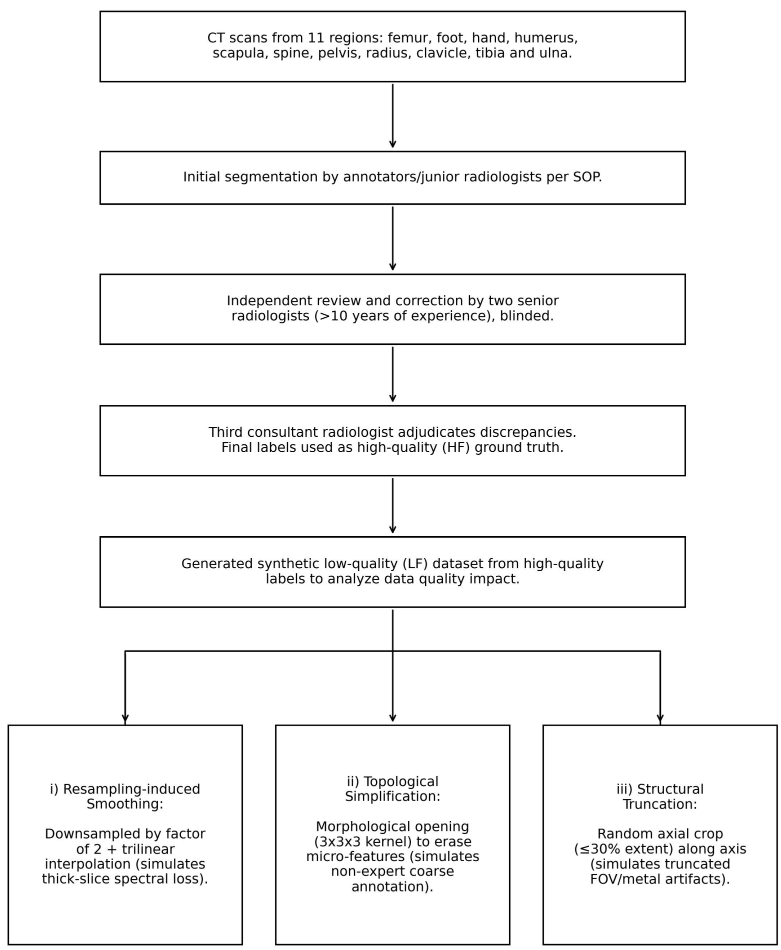
The network takes a 3D patch input (D × H × W) and outputs a binary segmentation map. The "Out Ch." denotes the number of output channels for each block. α represents the split ratio for the FFC global branch (or local, depending on definition).

Stage	Module	Output Channels	Resolution	Kernel Size / Stride	α (Split Ratio)	Configuration Details
Input	-	1	D×H×W	-	-	Raw intensity volume
Encoder 1	SCB	64	D×H×W	3×3×3 / 1	0.90	BN + ReLU; CPCA enabled
Downsample 1	MaxPool	64	D/2×H/2×W/2	2×2×2 / 2	-	-
Encoder 2	SCB	128	D/2×H/2×W/2	3×3×3 / 1	0.90	BN + ReLU; CPCA enabled
Downsample 2	MaxPool	128	D/4×H/4×W/4	2×2×2 / 2	-	-
Encoder 3	SCB	256	D/4×H/4×W/4	3×3×3 / 1	0.80	BN + ReLU; CPCA enabled
Downsample 3	MaxPool	256	D/8×H/8×W/8	2×2×2 / 2	-	-
Encoder 4	SCB	512	D/8×H/8×W/8	3×3×3 / 1	0.75	BN + ReLU; CPCA enabled
Downsample 4	MaxPool	512	D/16×H/16×W/16	2×2×2 / 2	-	-
Bottleneck	SDB	512	D/16×H/16×W/16	1×1×1 / 1	0.75	Parallel Spatial/Freq branches; Internal reduction ratio = 0.5
Decoder 4	Upsample + SCB	256	D/8×H/8×W/8	2×2×2 / 2	0.80	Transpose Conv upsampling
Decoder 3	Upsample + SCB	128	D/4×H/4×W/4	2×2×2 / 2	0.90	Deep Supervision Head 1 attached
Decoder 2	Upsample + SCB	64	D/2×H/2×W/2	2×2×2 / 2	0.90	Deep Supervision Head 2 attached
Decoder 1	Upsample + SCB	64	D×H×W	2×2×2 / 2	0.90	-

Stage	Module	Output Channels	Resolution	Kernel Size / Stride	α (Split Ratio)	Configuration Details
Output	Conv1x1	N_class	D×H×W	1×1×1 / 1	-	Final voxel-wise classification

Note:
SCB: Synergistic Convolution Block; SDB: Synergistic Decoupling Bottleneck.
Activation: All convolutional layers are followed by ReLU (inplace=False), except for the final segmentation head.
Normalization: Batch Normalization (3D) is used after each convolution.
CPCA Settings: The Contextual Path within CPCA utilizes multi-scale stripe convolutions with kernel sizes including 1×1×7, 1×1×11, 1×1×21, etc., to capture anisotropic context.

Supplementary Figure S7 Data generation and quality control pipeline.



Supplementary Figure S7: Schematic representation of the multi-stage expert adjudication process and the synthetic degradation protocol used to generate the High-Fidelity (HF) and Low-Fidelity (LF) datasets.